

Jake Schwartz

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SUMMARY

Software architect with 12+ years building backend services and real-time streaming data pipelines in Scala, with deep experience solving search problems in Elasticsearch. Has grown from engineer to engineering manager to architect, and now leads platform modernization and the adoption of LLM-powered and agentic systems, both in production and to streamline everyday engineering operations.

SKILLS

- **Languages:** Scala, Java, Python, TypeScript/JavaScript (Node.js)
- **Data & Streaming:** Apache Storm, Kafka (MSK), Akka/Pekko Streams, Elasticsearch, MongoDB, Redis, Databricks
- **AI & LLMs:** OpenAI and Databricks model integration, prompt engineering and caching, agentic workflows (Claude Code), computer vision (ResNet, YOLO)
- **Cloud & DevOps:** AWS (Lambda, ECS, SQS, S3, CodeArtifact), Pulumi, Docker, CircleCI, Datadog

EXPERIENCE

Signal Labs — Remote / San Francisco, CA

Senior Software Architect · Feb 2023 – Present

- Architected LLM-powered alerting that turns multi-signal detections (OCR, geolocation, deepfake analysis) into analyst-ready summaries. Analysts get actionable intelligence in place of raw signal noise.
- Designed a provider-agnostic LLM layer with automatic failover between OpenAI and Databricks-hosted models, prompt caching, and per-call cost attribution. The result is more reliable AI features with spend that can be tracked and kept under control.
- Architected multi-tenant pipelines on Storm, Kafka, and AWS that deliver CV-enriched images along with geo-referenced data, opening up a new revenue channel.
- Drove platform-wide modernization: Java 11 upgrades, a move from JFrog to AWS CodeArtifact across 35 repos, automated dependency management, and consolidated serverless jobs with Datadog anomaly monitoring. Security improved, operating costs came down, and maintenance got easier.
- Improved how Engineering leadership and stakeholders plan initiatives, set documentation standards, and built automated systems documentation that publishes to Confluence. Cross-team work became more predictable, documentation stays current as the platform changes, and new engineers onboard faster.

Data Engineering Manager · Sep 2020 – Feb 2023

- Managed and coached a data engineering team of 5–7 engineers, covering hiring, performance reviews, and career development to keep productivity and retention high.
- Defined and delivered quarterly OKRs tied to company objectives, turning strategic goals into measurable data initiatives and consistently hitting targets.
- Ran end-to-end Agile delivery (sprint planning, estimation, backlog grooming, retrospectives) and tracked velocity, cycle time, and PR turnaround to find bottlenecks and make delivery more predictable.
- Worked with product managers and business stakeholders to gather requirements and scope features, so solutions shipped on schedule and met quality and business needs.
- Stayed a hands-on technical lead on real-time Apache Storm and Kafka enrichment pipelines. Added new social and news sources, multilingual and regional enrichment, ML-based sentiment and entity extraction, and geolocation caching that improved coverage and cut third-party API costs.
- Standardized the Scala platform by upgrading services to Scala 2.13, moving MongoDB, AWS, and secrets access into a shared core library, and building a reusable CircleCI orb. Builds became consistent, duplicated code went away, and onboarding new services got faster.

Lead Software Engineer · May 2019 – Sep 2020

- Rebuilt the sprawling field-mapping and enrichment code as modular, reusable components, and retired legacy systems (Solr indexing, outdated sentiment and NLP integrations). The codebase got smaller and much easier to extend to new data sources.
- Replaced legacy services with Akka Streams and AWS Lambda, which cut infrastructure overhead and made processing jobs simpler to scale and operate.

- Designed and built large-scale data processing on Apache Storm, Kafka, Elasticsearch, and Redis, including real-time spike detection for early-warning alerts, story deduplication, and automated data-compliance processing, to support platform growth and customer requirements.
- Created a shared core Scala library and standardized services on a common web framework (Finatra) with consistent MongoDB, AWS, and Datadog integrations. Duplicated code went away and new services became faster to build.
- Pushed engineering standards by improving documentation, raising code coverage, adding integration tests, and enforcing consistent code style, which made releases more reliable across teams.

Senior Software Engineer, Search & Analytics · Oct 2017 – May 2019

- Improved search on Elasticsearch through schema and query optimization, performance tuning, and a major-version migration to ES 5, with zero-downtime index swaps using aliases. Queries got faster and the cluster ran more reliably.
- Architected a real-time content tagging system on Elasticsearch percolator queries, kept in sync with MongoDB through Change Streams. Customers' monitoring rules started applying to incoming content within seconds, and a companion service retroactively retagged historical data whenever rules changed.
- Re-architected historical data import around SQS and Kafka, replacing legacy job orchestration. Large backfills became more scalable and fault-tolerant, and the work helped drive the move to MongoDB Atlas.
- Led a range of major product and engineering epics from design through delivery, mentored junior engineers, replaced legacy architecture, and improved documentation.
- Made the team more productive by speeding up CircleCI builds, adding code-coverage and static-analysis metrics, and customizing an SBT release plugin to standardize releases, all while working effectively as a remote team member.

Software Engineer · Sep 2015 – Oct 2017

- Developed and tested new features for the backend REST APIs that served analytics and search data to the web application.
- Delivered key features for the real-time Apache Storm data pipeline, including field mapping and enrichment for Twitter (GNIP), news, and licensed content sources, plus geolocation and author metadata parsing, to improve data quality and coverage.
- Designed and launched a historical data import service with job previews, volume estimates, and completion notifications. Customers could backfill years of data into their analysis.
- Helped migrate the platform's search layer from Solr to Elasticsearch, removing legacy code and dependencies, and wrote the first automated tests for the backend search API.
- Established the team's first CI/CD pipelines (CircleCI, Docker image builds, Artifactory publishing) and helped the Engineering team move to a more disciplined Agile and GitFlow release process.

Elbrys Networks — Portsmouth, NH

Software Engineer · Jan 2014 – Sep 2015

- Helped develop and architect a Scala backend for a cloud-managed networking product (with Tallac Networks) that collected streaming device data and automated operations.
- Scaled the system to support over a thousand access points communicating with the backend continuously.
- Ran MongoDB migrations, index optimization, and sharding, and fixed application and database performance issues to improve responsiveness.
- Shared DevOps duties to keep production systems healthy.

Software Engineering Intern · Summers, May 2010 – Aug 2013

- Helped build and launch the company's first iOS and Android fitness apps, plus mobile SDKs that sent health data from devices to a cloud aggregation platform.
- Integrated third-party health APIs, including Fitbit, Runkeeper, Wahoo Fitness, and Withings, and joined the company's first Agile team.

University of New Hampshire — Durham, NH

Teaching Assistant · Sep 2012 – Dec 2013

- Taught programming fundamentals and object-oriented design in Introduction to Computer Science I and II.

EDUCATION

University of New Hampshire · Durham, NH

- M.S., Computer Science · 2012 – 2013
- B.S., Computer Science · 2008 – 2012 · Magna Cum Laude, University Honors